Optional Extension Activity

Investigating an Underwater Earthquake

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The 2011 Tohoku earthquake

On 11 March 2011, a magnitude 9.1 earthquake occurred offshore of northeastern Japan, along the Japan Trench. The earthquake resulted from sudden movement along the subduction boundary between tectonic plates and generated a devastating tsunami.

The event was detected by seismic instruments around the world. Underwater acoustic recordings can provide additional information about earthquake-generated signals.

Did you know?

NOAA scientists detected both seismic and underwater acoustic arrivals from the 2011 Tohoku earthquake using distant hydrophones.



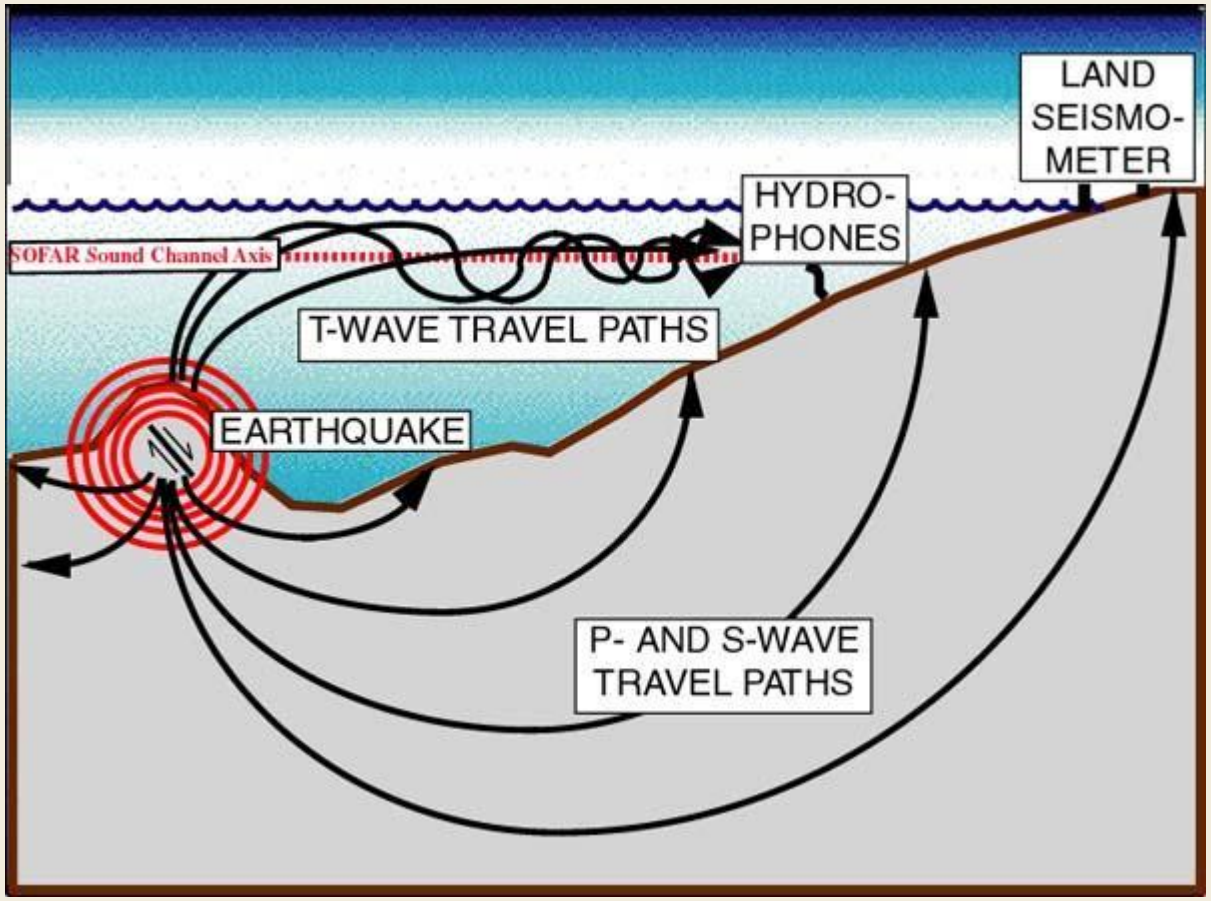
Photograph: Aftermath of the 2011 Tohoku earthquake and tsunami. Dylan McCord / U.S. Navy, via NOAA/NCEI.

Question: How can an earthquake beneath the seafloor produce sound that travels through the ocean?

How can an earthquake be heard underwater?

An underwater earthquake generates seismic waves and causes movement of the seafloor. Some of this energy is transferred into the water, generating acoustic waves. Under suitable ocean conditions, low-frequency acoustic signals can propagate over very long distances. Hydrophones record these pressure fluctuations, allowing researchers to investigate acoustic signals associated with earthquakes.

Key distinction: Seismometers detect ground motion, whereas hydrophones measure pressure fluctuations in water.



https://www.pmel.noaa.gov/eoi/nemo/education/curr_p2_07.html

Earthquake-generated acoustic T-waves are not tsunami waves. T-waves are sound waves that propagate through seawater, typically at approximately 1,500 m/s. Tsunamis are much longer-period gravity waves generated by large-scale water displacement.

The hydroacoustic recording

Dataset: Hydroacoustic recording of the 2011 Tohoku earthquake.

Data source: CTBTO International Monitoring System (IMS), hydroacoustic station HA11, Wake Island, Pacific Ocean.

Recording information

- Sampling frequency: 250 Hz
- Number of samples: 225,000
- Recording duration: 900 seconds (15 minutes)
- Nyquist frequency: 125 Hz
- Amplitude: Relative signal amplitude (digital units)

Source: CTBTO data used in Gomez and Kadri (2021).

Your challenge: Identify the main acoustic event and distinguish its characteristics from the background recording.

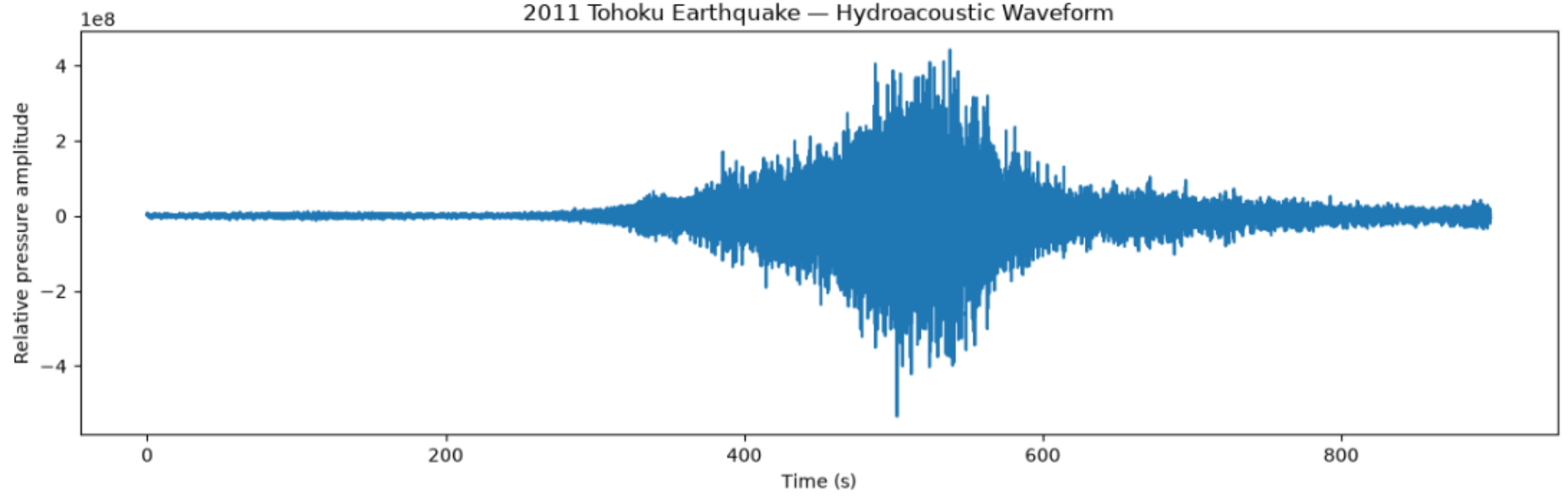
Notebook 3 — Investigating the 2011 Tohoku Earthquake.

Open the extension notebook and follow the four analyses in order:

- **Waveform:** Identify the main acoustic event and examine its amplitude and duration.
- **FFT:** Investigate which frequencies dominate the recording.
- **Spectrogram:** Explore how frequency content changes through time.
- **Filtering:** Compare the original signal with its filtered version.

Run each Python cell using Shift + Enter. Examine the resulting figures and answer the questions before continuing.

Remember: The recording is not calibrated to absolute acoustic pressure unless the appropriate instrument calibration has been applied.



Hydroacoustic signal associated with the 2011 Tohoku earthquake, recorded by CTBTO station HA11 and analysed in Gomez and Kadri (2021).
Gomez, B., & Kadri, U. (2021). *Scientific Reports*, 11, 23062. https://doi.org/10.1038/s41598-021-02483-w

What can hydroacoustics tell us about an earthquake?

Hydroacoustic recordings can reveal when acoustic energy arrives, how long a signal lasts and which frequencies are present. With suitable observations and analysis, hydroacoustic networks can also help detect and locate underwater seismic events.

Earthquake Analysis

1. **When does the main acoustic event occur, and how long does it last?**
   Identify the interval where the waveform amplitude increases substantially above background levels. Estimate its onset, peak and duration from the time axis.
2. **Which frequencies dominate the signal?**
   Identify the strongest peaks in the FFT spectrum. Use the spectrogram to determine whether these frequencies persist throughout the recording or are associated with the earthquake arrival.
3. **What changes after filtering?**
   Frequency components outside the selected passband are attenuated, making particular signal features easier to identify.
4. **Which analysis was most useful?**
   The spectrogram is particularly useful because it reveals both when the acoustic event occurs and how its frequency content changes over time. The waveform is useful for identifying its timing and amplitude.
5. **What cannot be determined from this recording alone?**
   Earthquake magnitude, fault slip, tsunami height and precise epicentral location cannot be reliably determined from this single recording without additional information or methods.

References

Scientific publications

- Gomez, B., & Kadri, U. (2021). Earthquake source characterization by machine learning algorithms applied to acoustic signals. *Scientific Reports, 11*, 23062. https://doi.org/10.1038/s41598-021-02483-w

Scientific and educational resources

- U.S. Geological Survey (USGS). *M 9.1 — 2011 Great Tohoku Earthquake, Japan*. https://earthquake.usgs.gov/earthquakes/eventpage/usc0001xgp
- NOAA Pacific Marine Environmental Laboratory (2011). *PMEL Scientists Hear Japan Earthquake*. https://pmel.noaa.gov/news-and-media/highlights/pmel-scientists-hear-japan-earthquake
- NOAA PMEL. *NeMO Curriculum — Oceanic Earthquakes*. https://www.pmel.noaa.gov/eoi/nemo/education/curr_p2_07.html

Figure and media credits

- Slide 2: Photograph of the aftermath of the 2011 Tohoku earthquake and tsunami. Dylan McCord / U.S. Navy, via NOAA/NCEI.
- Slide 3: Diagram illustrating seismic and hydroacoustic wave propagation. NOAA PMEL, NeMO educational curriculum.
- Slide 5: Hydroacoustic waveform of the 2011 Tohoku earthquake, generated from the dataset used in Gomez and Kadri (2021).

Dataset acknowledgement

- Slides 4–5: Hydroacoustic signal associated with the 2011 Tohoku earthquake, recorded by CTBTO station HA11 and analysed in Gomez and Kadri (2021).